

EXPERIENCE

Carnegie Mellon University Robotics Institute

Robotics Engineer (Biorobotics Lab)

Pittsburgh, PA

March 2026 - Current

TypeScript React Next.js Docker QuestDB SQLite

- Software lead on an industry sponsored research project to develop a modular IOT-enabled sensor network for digitizing medical gas pipeline systems in hospitals.
- Responsible for system architecture design, implementation of data collection and storage pipeline, and end-user friendly web dashboard for monitoring and alerting.

Took a break to explore interests like robotics and work on projects, including a full-sized humanoid(-ish) robot for more details and photos, please check out aliuliu121.github.io

Uber

Frontend Software Engineer (Unified Account Management Team)

Seattle, WA

September 2023 - July 2024

TypeScript JavaScript React GraphQL gRPC

- Worked on migrating Uber's fragmented account management to an unified service, providing a consistent user experience across Uber's 5+ apps and web clients.
 - Revamped existing account update flows, improved upon security features like 2FA enrollment, and authored new features like 3P data access management.
- Improved frontend reliability by advocating for and leading biweekly reliability meetings, writing the on-call runbook to provide a detailed plan of action to follow when things go wrong, and enhancing service alerting.

Backend Software Engineer (Product Equity Team)

July 2022 - September 2023

Go gRPC Thrift

- Architected and built Uber's automated name validation service, supplementing the existing update flow to reduce the volume of name change tickets sent for manual review. This has resulted in an estimated annual savings of \$500K on customer support costs in the US alone.

Software Engineer Intern (Cadence Team)

May 2021 - August 2021

Go Cadence Cassandra

- Developed an alternative DB-backed config system to the current file-based one for open source Cadence users.

Amazon Web Services

Seattle, WA (Remote)

Software Development Engineer Intern (CloudFormation Team)

June 2020 - August 2020

Java GraphQL AWS

- Designed a query engine for users to build complex queries using GraphQL against their CloudFormation data.

EDUCATION

Carnegie Mellon University

Bachelor of Science in Computer Science | Minor in Robotics

Pittsburgh, PA

September 2018 - May 2022

Relevant Courses Robot Kinematics and Dynamics, Introduction to Feedback Control Systems, Distributed Systems, Principles of Software Construction, Algorithm Design and Analysis, and Software Foundations of Security and Privacy

SKILLS

Programming Languages TypeScript JavaScript Python Go C/C++ Java

Frameworks and Technologies React Next.js Docker QuestDB SQLite GraphQL SQL ROS2

Other CAD 3D Printing Embedded Development Videography